

SUNAY BHATTACHARJEE

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EDUCATION

B.Tech in Computer Science and Engineering, NIT Surat 2024 – 2028
CGPA: 8.96/10 (after 4th semester)

TECHNICAL SKILLS

Programming Languages C++, Python, JavaScript	Core CS — Proficient Data Structures and Algorithms, OOPs, DBMS, Machine Learning Basics, Operating System
Web Technologies React JS, Django, Express JS, Node JS	Databases MySQL, Mongo DB, Postgres SQL
Tools Git, GitHub, Linux, VS Code, Jupyter	

PROJECTS

Emergency Truth Engine – Low Bandwidth Emergency Response System 📄,
Next.js, TypeScript, Dexie (IndexedDB), Supabase, Python, PyTorch, CLIP, DETR, RoBERTa, Xception

- Built an offline-first PWA for emergency reporting in low-connectivity environments with local storage using IndexedDB and automatic synchronization when connectivity becomes available.
- Implemented ML-powered emergency prioritization and authenticity verification using CLIP, DETR, RoBERTa, and Xception models for image, text, and deepfake analysis.
- Designed a scalable backend with Supabase and REST APIs, enabling multi-channel alert delivery, geolocation tracking, and fail-safe emergency escalation.
- Developed real-time analytics dashboards and compression pipelines to support reliable low-bandwidth communication and improve emergency response efficiency.

Algo Trading Bot 📄, *Machine Learning with Scikit-Learn, Python, Streamlit, yfinance*

- Built a hybrid algorithmic trading bot using Python, Scikit-learn, Streamlit, and yfinance, combining technical indicators with ML-based trend prediction and scalping strategies.
- Backtested and paper traded strategies on historical market data, incorporating risk management and performance evaluation metrics.
- Won the Genesis Competition for developing this ML-driven trading system.

Aqua-Verse Explorer 📄, *React.js, Next.js, Django REST Framework, PostgreSQL, Tailwind CSS*

- Built a full-stack marine exploration platform with an AI chatbot, community forum, and donation system.
- Developed secure REST APIs using Django REST Framework with JWT authentication and PostgreSQL.
- Designed a responsive React/Next.js frontend with seamless backend integration.

ACHIEVEMENTS & LEADERSHIP

- Achieved Codeforces Pupil rating and solved 500+ DSA problems across LeetCode, Codeforces, CodeChef, and CSES.
- Winner of the Genesis Competition for developing an ML-driven Algorithmic Trading Bot.
- Executive Member, ACM-NIT Surat, contributing to technical events and community initiatives.
- Representative, NEXUS NIT Surat, assisting in student engagement and departmental activities.